



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

DISTRIBUTED DATABASES PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Distributed Systems

TOTAL: 10 QUESTIONS

Q1 What is Distributed Databases and what problem in Distributed Systems does it solve?

Q6 How does Distributed Databases handle reliability and high availability?

Q2 What are the core components or concepts in Distributed Databases?

Q7 What observability does Distributed Databases provide out of the box?

Q3 How does Distributed Databases compare to its main alternatives?

Q8 What are common performance bottlenecks in Distributed Databases?

Q4 How do you get started with Distributed Databases for a new project?

Q9 What security best practices apply when running Distributed Databases?

Q5 What configuration options most affect Distributed Databases's behavior in production?

SSO / IdP

Q10 What mistakes do teams commonly make when adopting Distributed Databases?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Distributed Databases is a tool or platform

Mitigation

Distributed Databases is a tool or platform that automates or simplifies a specific concern in the Distributed Systems domain, reducing manual effort and operational complexity for engineering teams.

A6 Distributed Databases provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

Availability

A2 Distributed Databases has key building blocks that

Primitives

Distributed Databases has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Distributed Databases exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

Endpoint

A3 Distributed Databases trades off simplicity against feature richness differently than its alternatives. Choose Distributed Databases when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

Feature

A8 Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

Configuration

A4 Install Distributed Databases via its package manager

Gotchas

Install Distributed Databases via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Distributed Databases with the principle of

Assessment

Run Distributed Databases with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

Concurrency

A10 Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.

Documentation